Ryan Liu

438 N California Ave, Palo Alto, CA 94301 U.S. Citizen

ryanliu@princeton.edu ryanchenliu theryanl theryanl.github.io

Education

Princeton University Princeton, NJ
PhD in Computer Science Aug 2023 – Present

Carnegie Mellon University Pittsburgh, PA
Fifth Year MS in Computer Science Aug 2022 – May 2023
BS in Computer Science, College and University Honors Aug 2018 – May 2022

Research Experience

Princeton University (Advisors: Thomas L. Griffiths, Andrés Monroy Hernandez) Aug 2023 – Present

I study how large language models align with the patterns underlying human behavior. I also investigate applications surrounding the various representations of people that these models afford.

Carnegie Mellon University (Advisor: Nihar B. Shah) Dec 2019 – May 2022

I studied how to improve scientific peer review, addressing both reviewer biases (anchoring, citation bias) and improving reviewer-paper match quality. Results have been employed or informed the design at multiple top conferences.

Selected Publications

1. **Ryan Liu***, Jiayi Geng*, Addison J. Wu, Ilia Sucholutsky, Tania Lombrozo, and Thomas L. Griffiths. **"Mind Your Step (by Step): Chain-of-Thought can Reduce Performance on Tasks where Thinking Makes Humans Worse"** Preprint, 2024.
2. **Ryan Liu***, Jiayi Geng*, Joshua C. Peterson, Ilia Sucholutsky, and Thomas L. Griffiths. **"Large Language Models Assume People are More Rational than We Really are"** Preprint, 2024.
3. **Ryan Liu***, Theodore R. Sumers*, Ishita Dasgupta, and Thomas L. Griffiths. **"How do Large Language Models Navigate Conflicts between Honesty and Helpfulness?" Oral**, International Conference on Machine Learning 2024.
4. **Ryan Liu**, Howard Yen, Raja Marjieh, Thomas L. Griffiths, and Ranjay Krishna. **"Improving Interpersonal Communication by Simulating Audiences with Language Models"** Preprint, 2023.
5. Yihan Cao*, Shuyi Chen*, **Ryan Liu***, Zhiruo Wang, and Daniel Fried. **"API-Assisted Code Generation for Question Answering on Varied Table Structures"** EMNLP 2023.
6. **Ryan Liu** and Nihar B. Shah. **"ReviewerGPT? An Exploratory Study on Using Large Language Models for Paper Reviewing"** Oral, AAAI SDU 2024.
7. Tongshuang Wu, Haiyi Zhu, Maya Albayrak, Alexis Axon, Amanda Bertsch, Wenxing Deng, Ziqi Ding, Bill Guo, Sireesh Gururaja, Tzu-Sheng Kuo, Jenny T Liang, **Ryan Liu**, Ihita Mandal, Jeremiah Milbauer, Xiaolin Ni, Namrata Padmanabhan, Subhashini Ramkumar, Alexis Sudjianto, Jordan Taylor, Ying-Jui Tseng, Patricia Vaidos, Zhijin Wu, Wei Wu, Chenyang Yang. **"LLMs as Workers in Human-Computational Algorithms? Replicating Crowdsourcing Pipelines with LLMs"** Preprint, 2023.
8. **Ryan Liu**, Steven Jecmen, Fei Fang, Vincent Conitzer, and Nihar B. Shah. **"Testing for Reviewer Anchoring in Peer Review: A Randomized Controlled Trial"** Preprint, 2023.
9. Steven Jecmen, Hanrui Zhang, **Ryan Liu**, Fei Fang, Vincent Conitzer, and Nihar B. Shah. **"Near-Optimal Reviewer Splitting in Two-Phase Paper Reviewing and Conference Experiment Design"** AAAI HCOMP 2022, Best Paper Honorable Mention.
10. Ivan Stelmakh, Charvi Rastogi, **Ryan Liu**, Shuchi Chawla, Federico Echenique, and Nihar B. Shah. **"Cite-seeing and Reviewing: A Study on Citation Bias in Peer Review"** PLoS ONE.
11. Steven Jecmen, Hanrui Zhang, **Ryan Liu**, Nihar B. Shah, Vincent Conitzer, and Fei Fang. **"Mitigating Manipulation in Peer Review via Randomized Reviewer Assignments"** NeurIPS 2020.

Work Experience

Assistant in Instruction, Princeton COS 350 Ethics in Computing Sep 2024 – Present

I taught precepts, developed homework material, graded assignments, and held office hours to support student learning.

ACM Conference on Economics and Computation Workflow Chair Nov 2022 – Mar 2023

I developed and implemented the algorithm for paper-reviewer matching at EC'23 and conducted various analyses.

Facebook AI/ML Software Engineering Intern May – August 2021

I built an LSTM using Pytorch on 200 GB irregular time series data to predict future eye gaze positions for the Oculus headset.

Teaching Assistant, CMU 15-112 Fundamentals of Programming Jul – Dec 2019

I mentored 10 students' individual projects, with 2 mentees scoring in the top 6 of 400+ students, and created and taught a quantum computing special interest lecture, with Rigetti sponsoring $5000 compute for a live demo.

Invited Talks/Presentations

University of Texas at Austin NLP Group Nov 2024

Mind Your Step (by Step): Chain-of-Thought can Reduce Performance on Tasks where Thinking Makes Humans Worse

Oral, International Conference on Machine Learning 2024 Jul 2024

How do Large Language Models Navigate Conflicts between Honesty and Helpfulness?

Oral, AAAI SDU Workshop 2024 Feb 2024

ReviewerGPT? An Exploratory Study on Using Large Language Models for Paper Reviewing

London Machine Learning Meetup (Role: Moderator) Aug 2023

Joon Sung Park — Generative Agents: Interactive Simulacra of Human Behavior

Data Skeptics Podcast Jul 2023

Automated Peer Review

Carnegie Mellon University Meeting of the Minds May 2022

Identifying Human Biases in Peer Review via Real-Subject Experiments

Carnegie Mellon University Meeting of the Minds May 2021

Improving Algorithmic Tools for Conference Peer Review Research

Carnegie Mellon University Fall Undergraduate Research Showcase Dec 2020

Creating Robustness within Conference Peer Review

Carnegie Mellon University Meeting of the Minds May 2020

Assignment Algorithms to Prevent Quid-Pro-Quo in Conference Peer Review

Academic Honors & Professional Services

Reviewer, International Conference on Learning Representations 2025
Reviewer, Conference on Human Factors in Computing Systems 2025
Reviewer, Conference on Neural Information Processing Systems Behavioral ML Workshop 2024
Reviewer, Conference on Neural Information Processing Systems 2024
Student Organizer, Decentralized Social Media Workshop 2024
Workflow Chair, ACM Conference on Economics and Computation 2023
National Science Foundation Research Experience for Undergraduates Grant 2021
Carnegie Mellon University Summer Undergraduate Research Apprenticeship 2020